

Tung (Tony) Nguyen

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EDUCATION

Vassar College

B.A. in Computer Science | GPA: 3.95

Poughkeepsie, NY

Expected May 2028

- *Fellowship:* Cornell Tech Break Through Tech AI/ML Fellow (May 2026 – Present)
- *Awards:* NYCWiC Top-2 Undergraduate Poster; ASA DataFest 2025 Best Data Visualization; MLH HackNights Best Gemini API
- *Coursework:* Computer Systems, Data Structures & Algorithms, Software Engineering, Machine Learning

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C, OCaml, SQL (Postgres)

Backend & Frameworks: FastAPI, Node.js, Next.js, React, Firebase, Redis, Drizzle ORM, REST APIs, PostgreSQL

Systems & Tools: Git, GitHub Actions, CI/CD, Docker, Playwright, BrowserStack, Jest, Vercel, PyTorch

EXPERIENCE

Software Engineer Research Assistant

Jun. 2026 – Present

The de Leeuw Lab

Poughkeepsie, NY

- Architecting a serverless backend on **Firebase** with **Cloud Functions**, **Firestore**, **Cloud Storage**, and Auth, supporting realtime data flows and multi-tenant application state.
- Reduced per-execution inference cost **79%** by redesigning a distributed **LangGraph** workflow with isolated task contexts, targeted routing, and an LLM-based evaluation layer.
- Built a **GitHub Actions** CI/CD pipeline using **Playwright**, BrowserStack, Jest, and OpenSSF Scorecard, resolving **10+** dependency and configuration issues across an open-source JavaScript ecosystem.
- Optimized an agentic backend service that transforms research specifications into runnable **jsPsych** applications through automated planning, code generation, testing, and deployment, decreasing runtime by 67%.

Software Engineer

Jun. 2026 – Aug. 2026

Develop for Good

Remote

- Designed and shipped **15** test-driven REST endpoints across **8** route modules using **Next.js**, **Drizzle ORM**, and **Neon PostgreSQL**, with Zod validation and role-based access control.
- Built **3** React pages and fixed a critical application-wide rendering bug while validating an anonymity-preserving API layer that isolates submitter identity from endpoint responses.

Software Engineer Intern

Jun. 2025 – Sep. 2025

FPT Information Systems (FIS)

Hanoi, Vietnam

- Reduced API data-refresh latency **40%** by redesigning **FastAPI** endpoint contracts and integration logic for near-real-time synchronization between an internal API and application state.
- Reduced duplicate records **35%** across a **500K+** row dataset by redesigning the backend data-merging workflow for an internal data-quality platform from ambiguous requirements through production.

Machine Learning Research Assistant

Jan. 2026 – May. 2026

Vassar College Earth Science

Poughkeepsie, NY

- Built a multi-sensor binary water-mask pipeline (**rasterio**, **scikit-image**, **scipy**) processing **Sentinel-2**, **Landsat**, and **Planet** imagery, using a GPU-free design to run on lab hardware without deep-learning framework dependencies.
- Validated pipeline output against **DeepWaterMap**, achieving **90%** pixel-level agreement through heatmap, threshold, and mask-cleaning analysis.

PROJECTS

Instagent | [Python](#), [Redis](#), [FastAPI](#) [Link](#)

2026

- Engineered a fault-tolerant distributed task-execution system with **Redis**-backed contract-net bidding, claim/ack/dead-letter workflows, per-role circuit breakers, and bounded reviewer escalation.
- Benchmarked a multi-agent pipeline against a single-agent baseline using a Redis decision log and FastAPI dashboard, showing the review gate pays off on long-form outputs while adding overhead on short ones, and isolating critic latency as the bottleneck.

OBRE | [Java](#), [Android](#), [OpenCV](#), [ONNX](#), [Firebase](#) [Link](#)

Feb. 2025 – Present

- Architected a modular **Java** Android vision pipeline using CameraX, YOLOv12 ONNX, and ML Kit OCR, separating camera ingestion, concurrent inference, and output processing to maintain responsive real-time execution.
- Designed asynchronous processing between camera-frame acquisition and on-device inference/TTS, coordinating independent workloads to prevent expensive vision operations from blocking the UI thread under variable input rates.